

Assignment 2: Skin cancer

Data exploration & assessment exercise

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PUBLISHED

September 23, 2023

*Info: Some of the codes are inspired from the
skin_cancer_classification.ipynb*

Short data description

ISIC 2016:

The International Skin Imaging Collaboration (ISIC) is an academia and industry partnership focused on digital skin imaging for melanoma detection. The ISIC archive contains over 150,000 clinical and dermoscopic images with metadata, including diagnosis, anatomical location, size, melanoma features, patient information, and segmentation masks.

PH2:

The PH2 dataset is a collection of skin images and masks that highlight specific areas that are interesting for medical purposes. There are also classification files (.xlsx, .txt) that describe the images based on certain skin characteristics. One can use this dataset to study and analyze skin lesions and other conditions.

Overview

In the assignment i will examine both datasets and see if there are any unusual things in the data, like mistakes or biases. I will also check the distribution for each type of diagnosis and i will look at the shapes and sizes of the images to see if there are any special patterns in the data. I will figure out if i can use both datasets together for a specific job. Once that's done, i will divide the PH2 dataset into two parts: one for training and another for testing how good it is. Lastly, i will load the PH2 dataset using keras.

Import Libraries

Data loading

Shape of ISBIISBI2016_ISIC_Part3B: (900, 2)

Shape of PH2 dataset: (200, 17)

	Image Name	Histological Diagnosis	Common Nevus	Atypical Nevus	Melanoma	Asymmetry\n(I
0	IMD003	NaN	X	NaN	NaN	0
1	IMD009	NaN	X	NaN	NaN	0
2	IMD016	NaN	X	NaN	NaN	0
3	IMD022	NaN	X	NaN	NaN	0
4	IMD024	NaN	X	NaN	NaN	0

The clinical diagnosis (Common Nevus, Atypical Nevus, Melanoma) and the colors (White, Red, Light-Brown, Dark-Brown, Blue-Gray, Black) are Marked with X if chosen and NaN if not. Mapping them to 0 and 1 respectively is done in the next cell.

	Image Name	Histological Diagnosis	Common Nevus	Atypical Nevus	Melanoma	Asymmetry\n(I
0	IMD003	0	1	0	0	0
1	IMD009	0	1	0	0	0
2	IMD016	0	1	0	0	0
3	IMD022	0	1	0	0	0
4	IMD024	0	1	0	0	0

Now i will loop through each column and count unique values occurrence.
To get a better overview of the PH2 dataset

Column: 'Image Name' (dtype: object) has 200 unique values:

```
Image Name
IMD003      1
IMD312      1
IMD243      1
IMD251      1
IMD254      1
..
IMD384      1
IMD385      1
IMD389      1
IMD390      1
IMD435      1
```

Name: count, Length: 200, dtype: int64

Column: 'Histological Diagnosis' (dtype: object) has 7 unique values:

Histological Diagnosis

0	159
Melanoma	26
Dysplastic Nevus	5
Lentigo Maligna	5
Intradermal Nevus	2
Nodular Melanoma	2
Blue Nevus	1

Name: count, dtype: int64

Column: 'Common Nevus' (dtype: int64) has 2 unique values:

Common Nevus

0	120
1	80

Name: count, dtype: int64

Column: 'Atypical Nevus' (dtype: int64) has 2 unique values:

Atypical Nevus

0	120
1	80

Name: count, dtype: int64

Column: 'Melanoma' (dtype: int64) has 2 unique values:

Melanoma

0	160
1	40

Name: count, dtype: int64

Column: 'Asymmetry

(0/1/2)' (dtype: int64) has 3 unique values:

Asymmetry\n(0/1/2)

0	117
2	52
1	31

Name: count, dtype: int64

Column: 'Pigment Network

(AT/T)' (dtype: object) has 2 unique values:

Pigment Network\n(AT/T)

AT	116
T	84

Name: count, dtype: int64

Column: 'Dots/Globules

(A/AT/T)' (dtype: object) has 3 unique values:

Dots/Globules\n(A/AT/T)

A	87
AT	59
T	54

Name: count, dtype: int64

```
Column: 'Streaks'
(A/P)' (dtype: object) has 2 unique values:
Streaks\n(A/P)
A    170
P     30
Name: count, dtype: int64
```

```
Column: 'Regression Areas'
(A/P)' (dtype: object) has 2 unique values:
Regression Areas\n(A/P)
A    175
P     25
Name: count, dtype: int64
```

```
Column: 'Blue-Whitish Veil'
(A/P)' (dtype: object) has 2 unique values:
Blue-Whitish Veil\n(A/P)
A    164
P     36
Name: count, dtype: int64
```

```
Column: 'White' (dtype: int64) has 2 unique values:
White
0    181
1     19
Name: count, dtype: int64
```

```
Column: 'Red' (dtype: int64) has 2 unique values:
Red
0    190
1     10
Name: count, dtype: int64
```

```
Column: 'Light-Brown' (dtype: int64) has 2 unique values:
Light-Brown
1    139
0     61
Name: count, dtype: int64
```

```
Column: 'Dark-Brown' (dtype: int64) has 2 unique values:
Dark-Brown
1    156
0     44
Name: count, dtype: int64
```

```
Column: 'Blue-Gray' (dtype: int64) has 2 unique values:
Blue-Gray
0    162
1     38
Name: count, dtype: int64
```

```
Column: 'Black' (dtype: int64) has 2 unique values:
Black
```

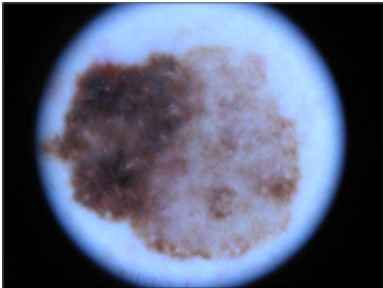
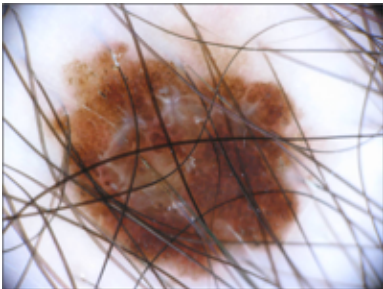
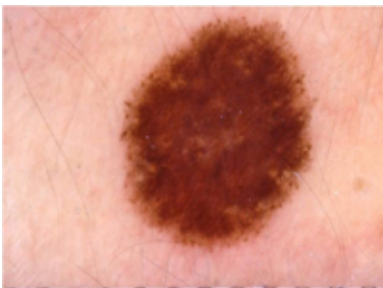
```
0    158
1     42
Name: count, dtype: int64
```

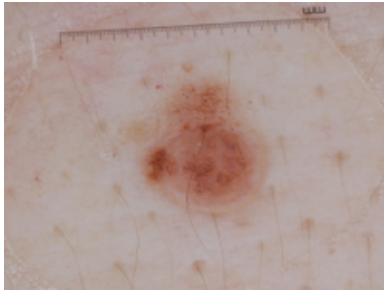
Task 1: Explore and evaluate both datasets

Biasing artifacts

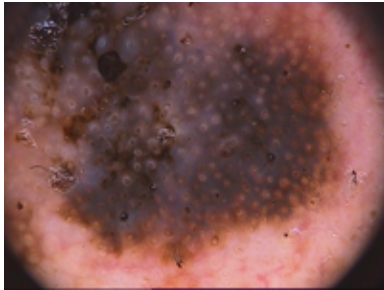
I have viewed at least 20-30 random samples from the PH2 dataset images and I can see that the images are dermoscopy images and all have good lighting with very less reflection. If hair follicles are present or dry skin surface, then the pictures are also focused on that part which leads to some small biases. All have a small black border around the edges but not that significant or which would lead to bad resolution of the image. The lesions on the skin are very good proportioned to the image and there are no blurred images. I also see no clinical markings or bandages.

The ISIC dataset has several biases. Only some of them are displayed here:

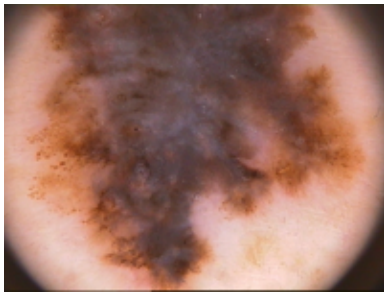
Images: isic10	Description
	Border
	Hair
	Ruler



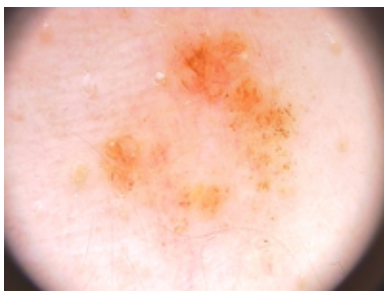
Scale/Ruler



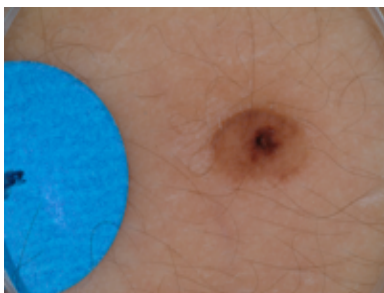
Zoom



Zoom



Lightning



Object

Lable Distribution

The PH2 dataset consists of 200 dermoscopic images, categorized into three classes: 80 Common Nevus, 80 Atypical Nevus, and 40 Melanomas. The dataset appears to have an imbalanced class distribution with a smaller number of melanomas compared to the other classes.

The ISBI2016 dataset consists of 900 training images and 379 test images for tasks such as lesion segmentation, dermoscopic feature detection, and melanoma classification. It also has a class imbalance.

Shape of ISBI2016_ISIC_Part3B: (900, 2)

Shape of PH2 dataset: (200, 17)

Common Nevus

0 120

1 80

Name: count, dtype: int64

Atypical Nevus

0 120

1 80

Name: count, dtype: int64

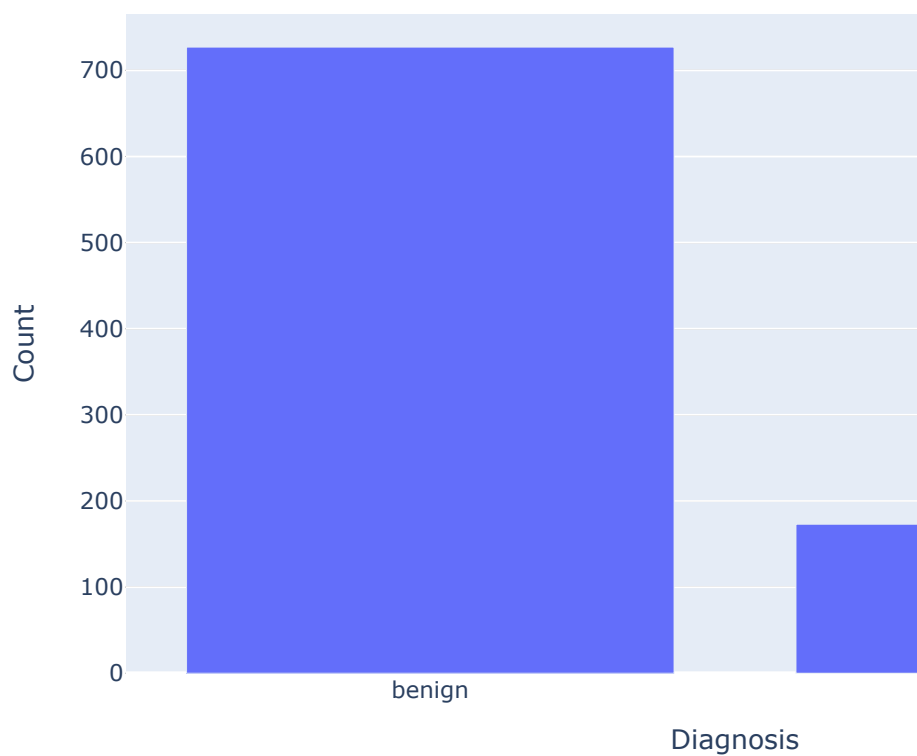
Melanoma

0 160

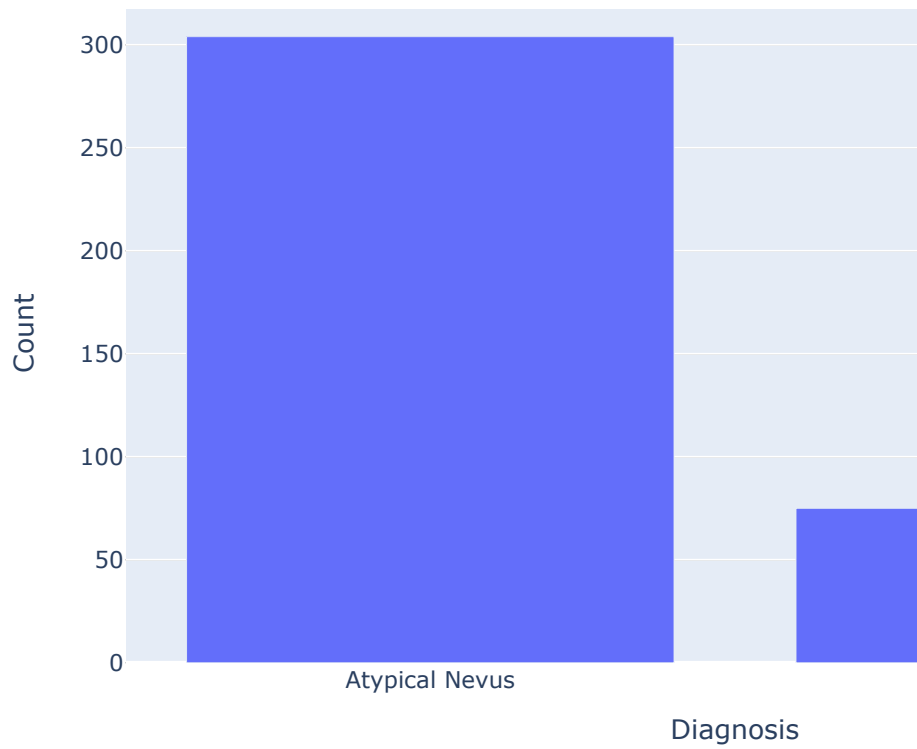
1 40

Name: count, dtype: int64

Class Distribution



Class Distribution



Aspect Ratios and Image Dimensions

Image Width Distribution

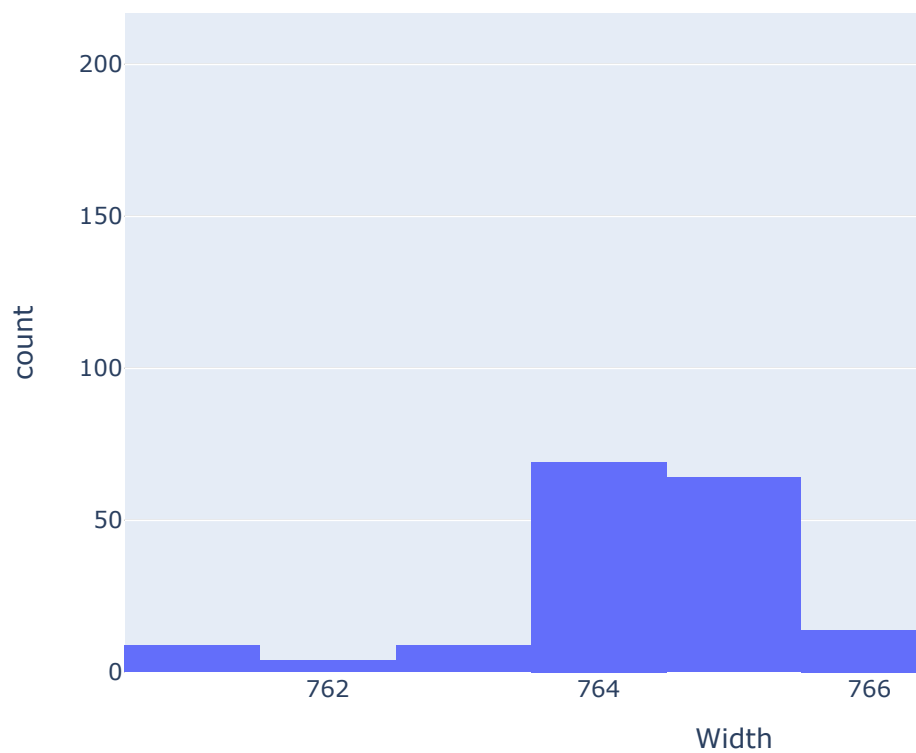


Image Height Distribution

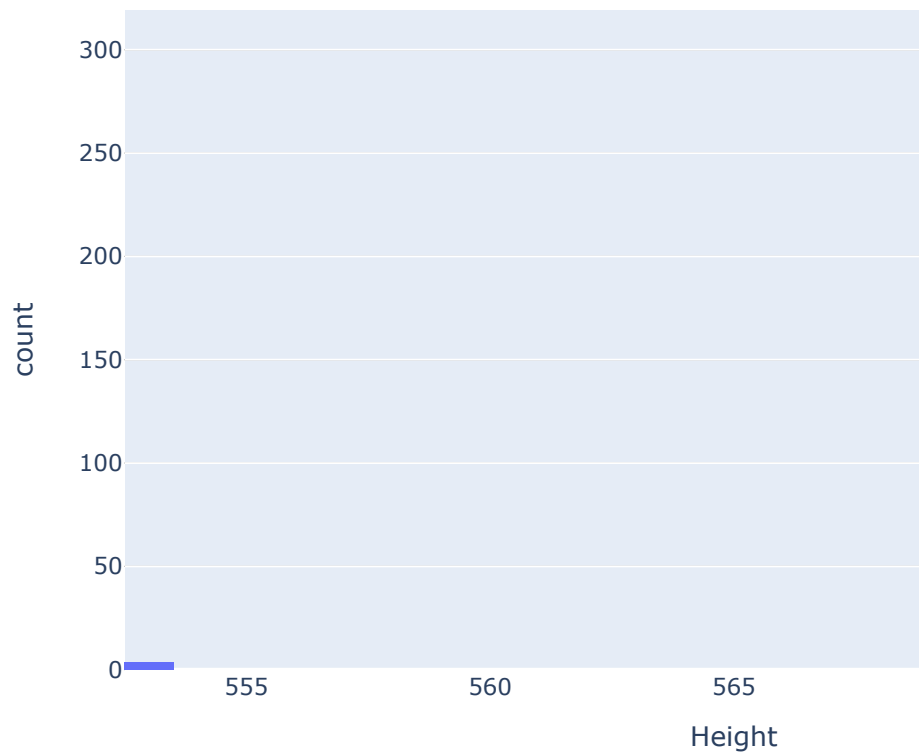


Image Aspect Ratio Distribution

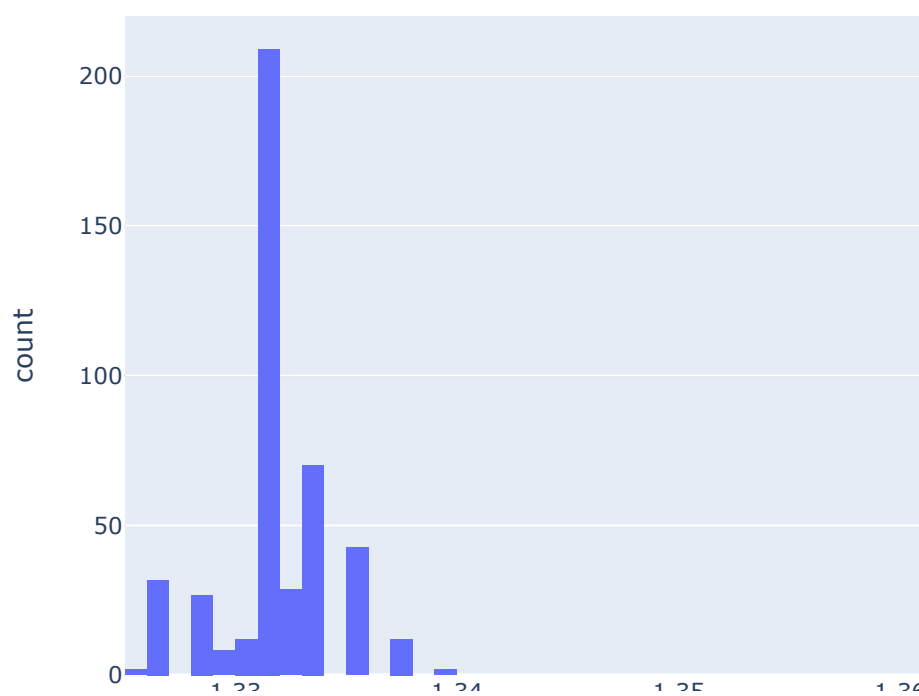




Image Width Distribution

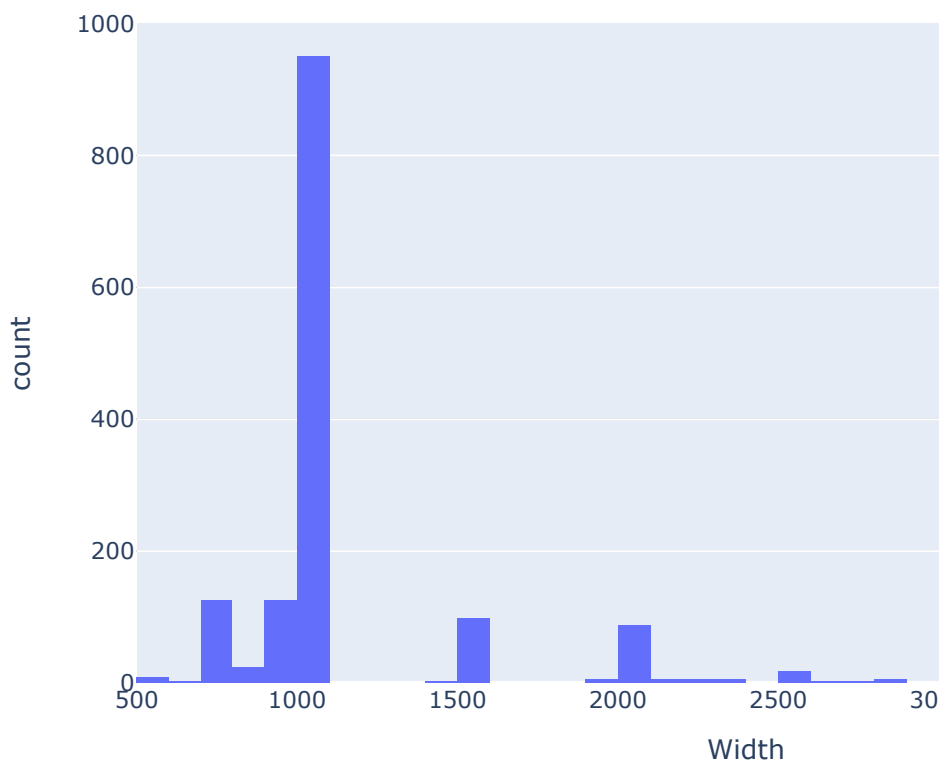
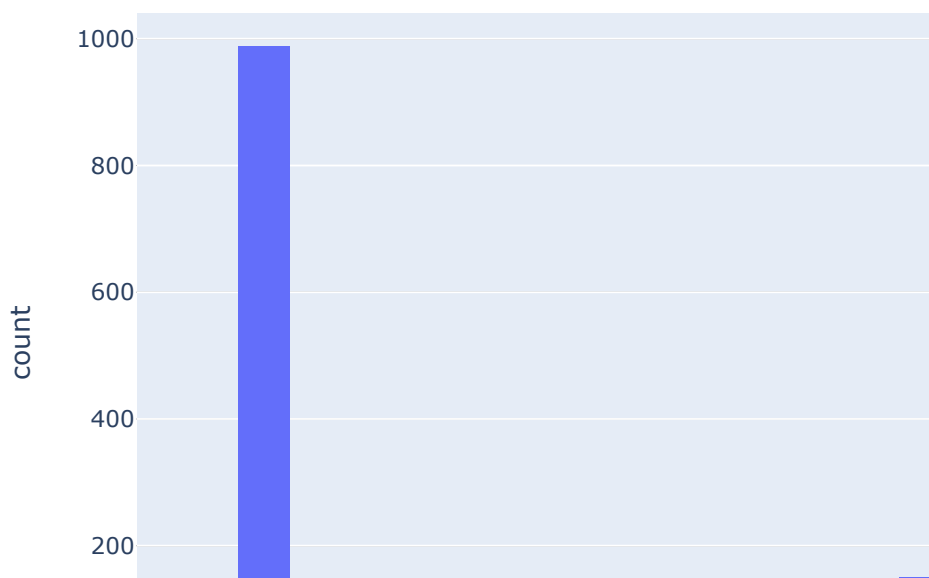


Image Height Distribution



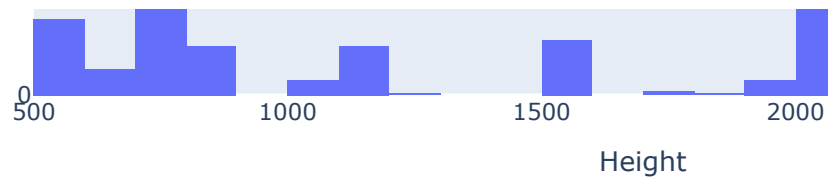


Image Aspect Ratio Distribution

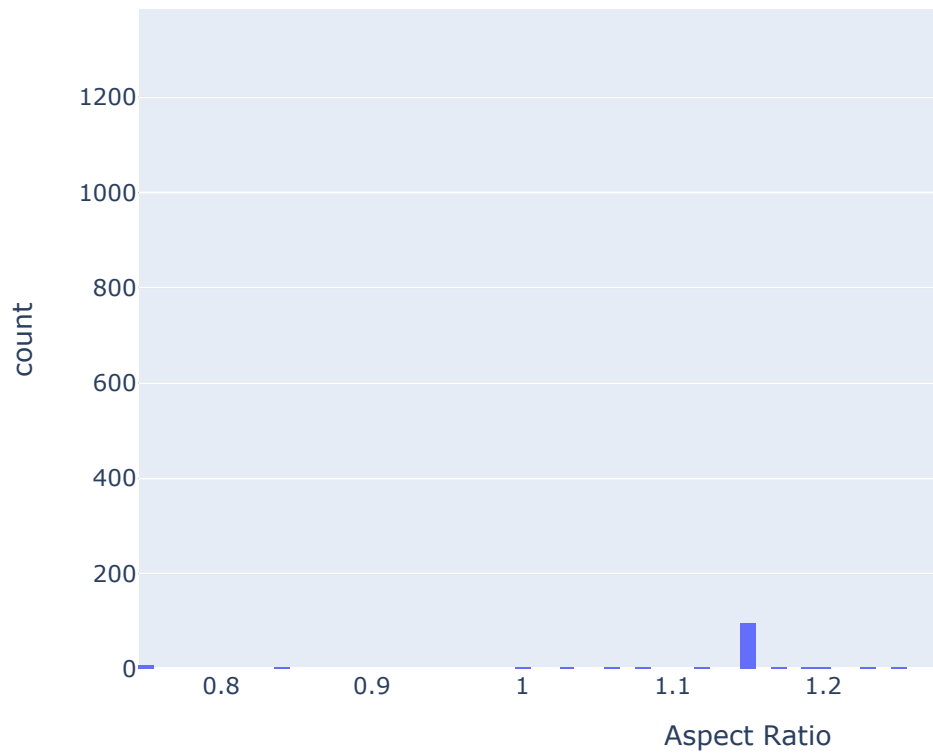
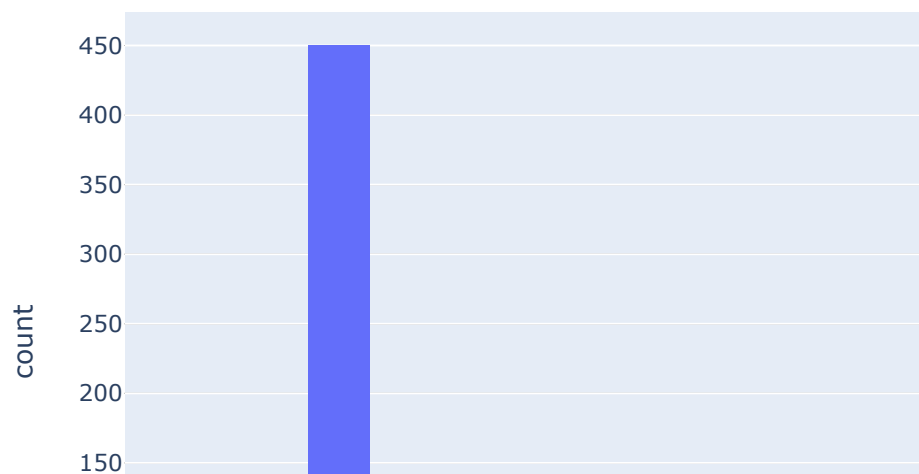


Image Width Distribution



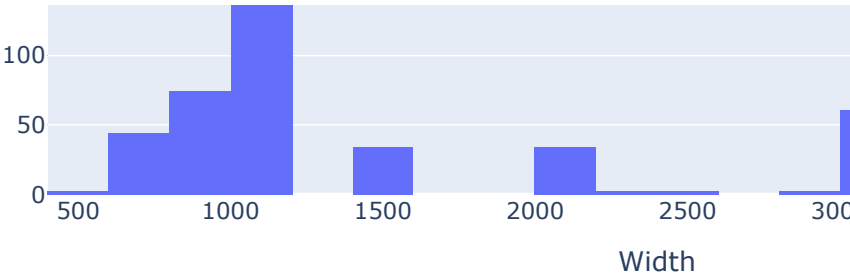


Image Height Distribution

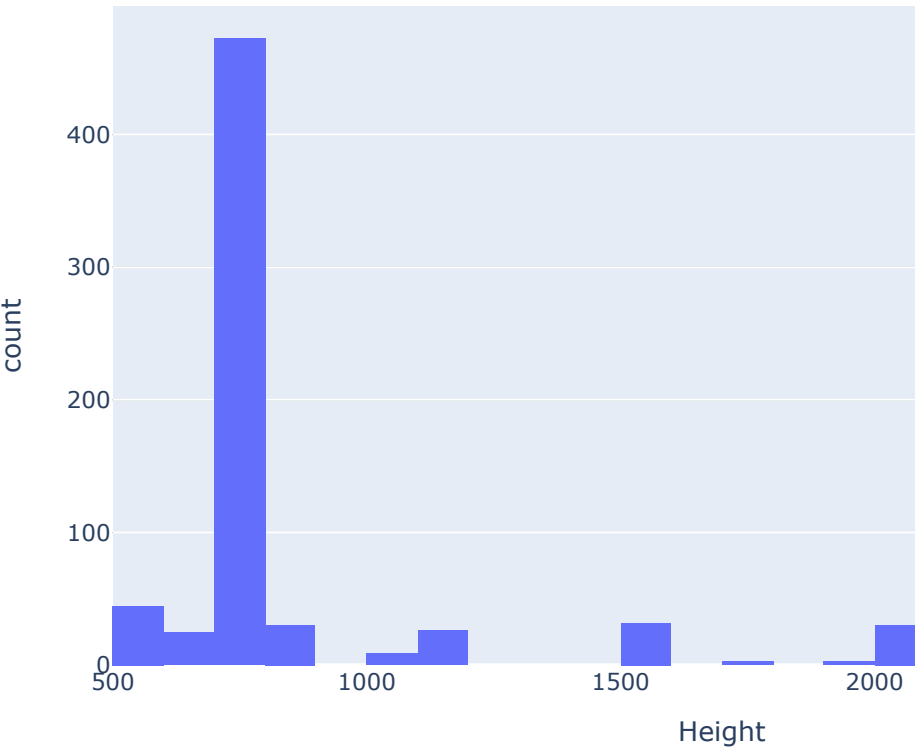
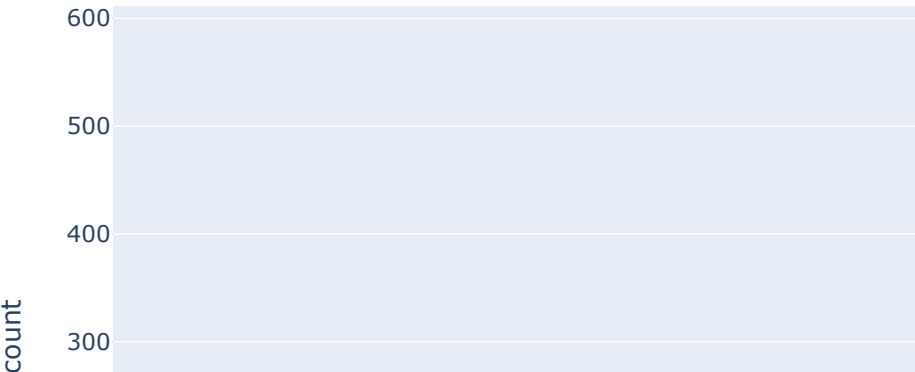
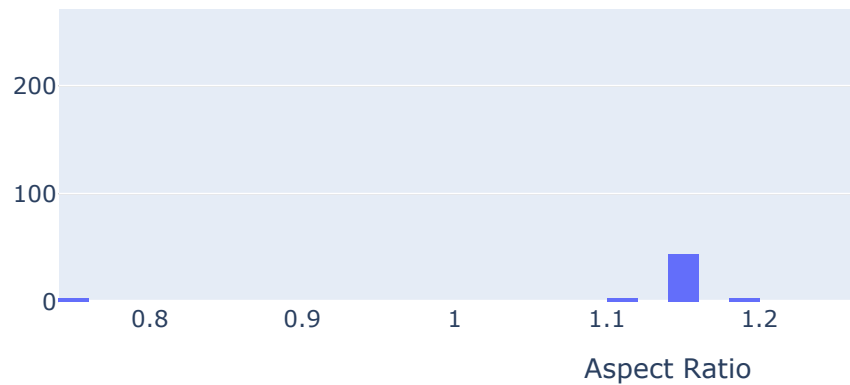


Image Aspect Ratio Distribution





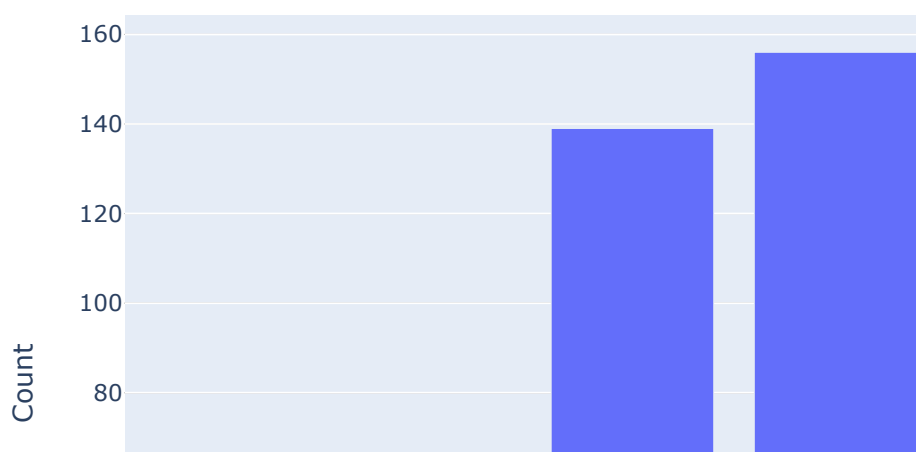
The width in the ISIC dataset has a range from 400–4400 pixels and the height has a range from 500–2900 pixels. The aspect ratio is between 0.5 and 1.5. The PH2 dataset has a width range from 760–780 pixel and the height range from 552–578 pixels. The aspect ratio is 1.3. The aspect ratio is not that important for the classification task, but it is important for the segmentation task.

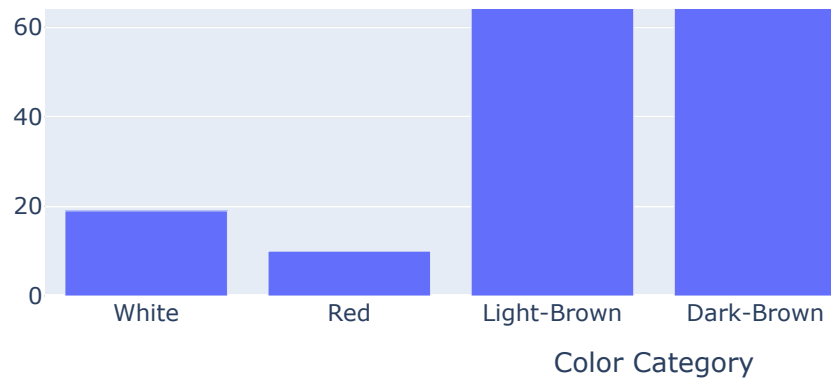
ISIC dataset contains images with a wider range of sizes than the PH2 dataset. The PH2 dataset has more consistent image dimensions and aspect ratios.

Patterns in Data

As shown earlier there are class imbalances in both datasets regarding the feature diagnosis. The ISIC dataset has a wide range of image dimensions and while the PH2 is consistent. In the PH2 dataset there are image segmentation which are better than in the ISIC dataset. For example the PH2 dataset identifies regions of interest (ROIs) within skin lesions. Analyzing pixel intensity or color distribution within these ROIs can help in the diagnosis of melanoma. The PH2 dataset has color profiles which are feature engineered from images.

Combined Distribution of Colors





Compatibility of Datasets

While it is technically possible to use multiple datasets together, the PH2 dataset and the ISIC dataset are not compatible with each other. The PH2 dataset has a different image format than the ISIC dataset. The PH2 dataset has a .bmp format and the ISIC dataset has a .jpg format and .png. The shapes of them are also different, in terms of number of images and number of feature columns. The ISIC has 2 and PH2 17. Also regarding the diagnosis labeling in the ISIC dataset it is between (Benign and Malignant) and the PH2 dataset has three different diagnosis (Common Nevus, Atypical Nevus, Melanoma). Which is important if the machine learning project goal differs.

In all together they are technically compatible, but not in terms of the labeling and the image format.

PH2 dataset Train/Test Split

Task 2: Prepare Keras/PyTorch Data Loaders for PH2 Dataset

This part has many similarities with the `skin_cancer_classification.ipynb`. I have used the same code and changed it to fit the PH2 dataset. Thanks for providing the code and documentation.

Data Loading

```
Found 160 files belonging to 3 classes.  
Using 128 files for training.  
Found 160 files belonging to 3 classes.  
Using 32 files for validation.
```

Found 40 files belonging to 3 classes.

Data Preprocessing and Augmentation

Visualize the batches
